Technical Supplement

SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse-Reward Hindsight Experience Replay

Anonymous Submission

Overview. This supplement gives a complete mathematical specification and pseudocode for SABER, followed by experimental settings, seed-level primary results, sensitivity and scope controls, support-signal diagnostics, runtime measurements, and an index of the anonymous code-and-data artifacts.

Working-draft completion status. One result slot remains *pending*: FPP DDPG Actor+Critic. No value is inferred from its incomplete run. This marker and paragraph must be removed after a complete, validated five-seed artifact replaces the remaining slot.

1 Complete SABER Specification

1.1 Scope, Problem Setting, and Notation

SABER is a critic-loss allocation rule for replay-based off-policy actor–critic learning. It is neither a replay sampler nor a network module. Consider a finite-horizon goal-conditioned MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{G}, P, r, \gamma, H).$$

The agent receives observation o_t, desired goal g, and network input $x_t = (o_t, g)$. The achieved-goal map is $m : \mathcal{S} \to \mathcal{G}$. The sparse reward used in the experiments is

$$r(s_t, a_t, s_{t+1}, g) = \begin{cases} 0, & d(m(s_{t+1}), g) \le \epsilon_g, \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

Replay stores the executed action a_i^r. We reserve a_i^π for the detached current-policy action used only by SABER's discrete support query. There are $n_Q = 1$ critics for DDPG and $n_Q = 2$ for TD3 and SAC.

1.2 Future-Goal HER Batch and Unchanged Bellman Targets

Let an episode contain states $s_0, \dots, s_T$ and transitions indexed by $t = 0, \dots, T-1$. When transition t is selected for relabeling and has an eligible later stored transition, the implemented future-goal convention samples

$$\begin{aligned} u &\sim \mathrm{Unif}\{t+1, \dots, T-1\}, \\ \widetilde{g}_i &= m(s_{u+1}), \\ \widetilde{r}_i &= r(s_t, a_t, s_{t+1}, \widetilde{g}_i). \end{aligned} \tag{2}$$

If the eligible set is empty, or if relabeling is not selected, the original desired goal and its corresponding reward are retained. Thus the nominal relabeling probability is $p_{\mathrm{HER}} = 0.8$, while the realized fraction can be slightly lower near an episode boundary. The relabeled network inputs are

$$x_i = (o_i, \widetilde{g}_i), \qquad x_i' = (o_i', \widetilde{g}_i).$$

Let $\chi_i \in \{0, 1\}$ be the bootstrap mask. Successful relabeled transitions and time-limit transitions use $\chi_i = 1$; genuine terminating transitions use $\chi_i = 0$. For backbone A, define

$$y_i^{\mathsf{A}} = \widetilde{r}_i + \gamma \chi_i V_i^{\mathsf{A}}(x_i'), \tag{3}$$

where target values are evaluated without gradient. Writing barred parameters for target networks, the implemented continuation values have the standard forms

$$V_i^{\mathrm{DDPG}} = Q_{\bar{\phi}_1}(x_i', \mu_{\bar{\theta}}(x_i')), \tag{4}$$

$$\begin{aligned} \varepsilon_i &\sim \mathrm{clip}\big(\mathcal{N}(0, \sigma_{\mathrm{targ}}^2 I), -c, c\big), \\ \widetilde{a}_i' &= \mathrm{clip}_{\mathcal{A}}(\mu_{\bar{\theta}}(x_i') + \varepsilon_i), \\ V_i^{\mathrm{TD3}} &= \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', \widetilde{a}_i'), \end{aligned} \tag{5}$$

$$\begin{aligned} a_i' &\sim \pi_\theta(\cdot \mid x_i'), \\ V_i^{\mathrm{SAC}} &= \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', a_i') - \alpha \log \pi_\theta(a_i' \mid x_i'). \end{aligned} \tag{6}$$

SABER does not change any quantity in Eqs. (2)–(6). For critic k, the replay-action Bellman residual remains

$$\delta_{ik} = Q_{\phi_k}(x_i, a_i^r) - y_i^{\mathsf{A}}. \tag{7}$$

Under ordinary uniformly sampled HER, every squared residual has explicit coefficient one.

1.3 Current-Policy Local Support

For each relabeled sample, SABER forms the full discrete observation–goal–horizon context

$$c_i = C(o_i, \widetilde{g}_i, h_i), \tag{8}$$

where h_i is the remaining horizon. A lower-dimensional retrieval key $C^{\mathrm{ret}}(o_i, h_i)$ is used only to query the replay index. Let $\mathbf{J}_i^{\mathrm{ret}}$ be the resulting ordered list of currently valid, unique replay slots. Invalid entries and duplicate slot identifiers are removed in maintained index order, retaining the first valid occurrence. The bounded scan is

$$S_i = \mathrm{Tail}_{N_{\mathrm{scan}}}\left(\mathbf{J}_i^{\mathrm{ret}}\right), \qquad |S_i| \le N_{\mathrm{scan}}. \tag{9}$$

The tail refers to deterministic maintained index order, not guaranteed temporal recency after circular-buffer wraparound. The coarse key is only an acceleration device. Support eligibility is determined by exact component-wise equality of the full context:

$$I_i = \{j \in S_i : C(o_j, \widetilde{g}_i, h_j) = C(o_i, \widetilde{g}_i, h_i)\}. \tag{10}$$

The query's relabeled goal $\widetilde{g}_i$ is used on both sides; a candidate's originally stored desired goal is not used for matching. The sampled transition is not removed from consideration and may therefore occur in I_i.

For action dimension d, let ℓ_d and u_d be its environment bounds. SABER first maps the action to

$$z_d(a) = \mathrm{clip}\left(\frac{a_d - \ell_d}{u_d - \ell_d}, 0, 1\right) \tag{11}$$

and assigns the discrete component

$$b_{K,d}(a) = \min\{K - 1, \lfloor K z_d(a) \rfloor\}. \tag{12}$$

The multidimensional bin $b_K(a)$ is the tuple of these components; the implementation stores its equivalent base-K integer encoding. The local set of occupied replay-action bins is

$$\mathcal{U}_i = \left\{b_K(a_j^r) : j \in I_i\right\}, \qquad n_i = |I_i|. \tag{13}$$

The binary support indicator is

$$\psi_i = \mathbf{1}\{n_i \ge N_{\mathrm{local}}\}\, \mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}. \tag{14}$$

If retrieval returns no candidates, exact matching yields too few candidates, or the policy bin is unoccupied, then $\psi_i = 0$. Main SABER uses

$$K = 5, \qquad N_{\mathrm{scan}} = 512, \qquad N_{\mathrm{local}} = 4.$$

DDPG and TD3 use their deterministic actor output as a_i^π. SAC uses the deterministic squashed actor mean for this query only; its target and actor updates continue to use stochastic policy samples. No gradient passes through a_i^π, action binning, retrieval, or ψ_i.

Task family	Full context components	Coarse retrieval key
FetchPickAndPlace	Phase, hand–object distance, object–goal distance, object height, gripper gap, horizon	Hand–object distance, object height, gripper gap, horizon
DrawerOpen/Close	Phase, hand–handle distance, handle–goal distance, handle y, gripper state, horizon	Hand–handle distance, handle y, gripper state, horizon

Table 1: Task-specific support contexts. Candidate retrieval uses the coarse key, after which exact component-wise equality on the full discretized context defines the local replay set.

Component	FetchPickAndPlace thresholds	Drawer thresholds
Phase	approach, grasp, lift, place	approach, contact, opening/closing, open/closed
Hand distance	.03, .05, .08, .12, .20, .40	.03, .05, .08, .12, .20, .40
Goal distance	.05, .10, .20, .40, .70	.03, .05, .08, .12, .20, .40, .70
Task coordinate	object z: .43, .48, .55, .65, .75	handle y: $-.05, 0, .05, .10, .15, .25$
Gripper	gap: .02, .05, .08	state: $-.50, 0, .50$
Remaining horizon	$10, 25, 50, 100$	$10, 25, 50, 100, 200, 500$

Table 2: Discretization boundaries used to construct support contexts. Values are applied component-wise in environment units. The main method normalizes each action dimension to its action bounds and uses $K = 5$ bins per dimension, $N_{\mathrm{scan}} = 512$, $N_{\mathrm{local}} = 4$, and $\lambda_S = 1$.

For the Drawer phase definitions, let d_h be hand–handle distance and d_g be handle–goal distance. The terminal phase is open or closed when $d_g \le 0.05$; opening or closing requires $d_h \le 0.08$ and $0.05 < d_g \le 0.12$; contact requires $d_h \le 0.08$ and $d_g > 0.12$; remaining samples are approach. Phases use the relabeled training goal.

1.4 Mean-Normalized Critic-Only Objective

SABER assigns the raw coefficient

$$w_i^{\mathrm{raw}} = 1 + \lambda_S \psi_i \tag{15}$$

and normalizes it by the minibatch mean:

$$\bar{w}_B = \frac{1}{|B|} \sum_{j \in B} w_j^{\mathrm{raw}}, \qquad \Omega_i = \frac{w_i^{\mathrm{raw}}}{\max(\epsilon_w, \bar{w}_B)}. \tag{16}$$

We use $\lambda_S = 1$ and $\epsilon_w = 10^{-8}$. The safeguard is inactive because $w_i^{\mathrm{raw}} \geq 1$. If $p_B = |B|^{-1} \sum_i \psi_i$, then

$$\Omega_i = \begin{cases} \dfrac{1 + \lambda_S}{1 + \lambda_S p_B}, & \psi_i = 1, \\ \dfrac{1}{1 + \lambda_S p_B}, & \psi_i = 0. \end{cases} \tag{17}$$

Hence every sample remains in the update with a positive coefficient, the minibatch mean coefficient is one, and support-positive samples receive relative weight $1 + \lambda_S$ compared with support-negative samples. If all indicators in a minibatch are equal, then every normalized coefficient is exactly one.

The critic objective is

$$\mathcal{L}_Q^{\mathrm{SABER}} = \frac{1}{|B|} \sum_{i \in B} \Omega_i \sum_{k=1}^{n_Q} \delta_{ik}^2. \tag{18}$$

For TD3 and SAC, the same Ω_i multiplies both critic residuals. The main variant is critic-only:

$$\mathcal{L}_\pi^{\mathrm{SABER}} = \mathcal{L}_\pi^{\mathsf{A}}. \tag{19}$$

Mean-one normalization preserves the average explicit coefficient, but does not imply an unchanged numerical loss or gradient norm because support may correlate with residual magnitude.

1.5 End-to-End Update

Algorithm 1 states one complete learning update. The replay index is maintained as transitions enter or overwrite replay slots. Candidate retrieval supplies only a bounded list; exact full-context matching determines support eligibility.

Algorithm 1: One SABER update

Require: replay $\mathcal{D}$ and index $\mathcal{R}$; actor π_θ; critics $\{Q_{\phi_k}\}_{k=1}^{n_Q}$; target networks
Require: context maps C, C^{ret}; $K, N_{\mathrm{scan}}, N_{\mathrm{local}}, \lambda_S$

$B \leftarrow \mathrm{UniformSample}(\mathcal{D})$
Apply Eq. (2) where selected and eligible; recompute rewards.
for each relabeled sample $i \in B$ **do**
  Construct x_i, x_i', bootstrap mask χ_i, and detached y_i^{A} using Eqs. (3)–(6).
  $a_i^\pi \leftarrow \mathrm{sg}[\mathrm{QueryAction}_{\mathsf{A}}(x_i)]$
  $q_i^{\mathrm{ret}} \leftarrow C^{\mathrm{ret}}(o_i, h_i)$
  $\mathbf{J}_i^{\mathrm{ret}} \leftarrow \mathcal{R}[q_i^{\mathrm{ret}}]$
  Remove invalid and duplicate slots from $\mathbf{J}_i^{\mathrm{ret}}$.
  $S_i \leftarrow \mathrm{Tail}_{N_{\mathrm{scan}}}(\mathbf{J}_i^{\mathrm{ret}})$
  $I_i \leftarrow \{j \in S_i : C(o_j, \widetilde{g}_i, h_j) = C(o_i, \widetilde{g}_i, h_i)\}$
  $\mathcal{U}_i \leftarrow \{b_K(a_j^r) : j \in I_i\}$
  $\psi_i \leftarrow \mathbf{1}\{|I_i| \geq N_{\mathrm{local}}\}\mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}$
  $w_i^{\mathrm{raw}} \leftarrow 1 + \lambda_S \psi_i$
end for
$\Omega_i \leftarrow w_i^{\mathrm{raw}} / \max(\epsilon_w, |B|^{-1} \sum_{j \in B} w_j^{\mathrm{raw}})$ for every i.
Compute replay-action residuals using Eq. (7).
Update all critics with Eq. (18).
Apply the backbone's unchanged actor, temperature, delayed-update, and target-network rules.

The algorithm introduces no auxiliary target and never regresses the critic at a_i^π. It does not alter minibatch membership, HER relabeling, reward recomputation, exploration, network architecture, optimizer, or UTD. The indicator is a binary occupancy proxy, not an estimate of density, uncertainty, target accuracy, target variance, or TD-error rank. Finally, SABER cannot mark an action as locally supported unless its action bin is already represented by replay actions in the matched context.

2 Experimental and Reproducibility Details

2.1 Tasks, Cohorts, and Shared Training Settings

Tables 3–5 give the settings shared within each matched cohort. The primary comparison contains seven task–backbone cohorts and uses paired seeds 1–5. Drawer SAC is not a primary cohort because a complete matched HER–SABER pair was not collected.

Task	Horizon	Budget	Backbones	Exact primary grid
FetchPickAndPlace-v4	50	1M	DDPG, TD3, SAC	10k (DDPG/TD3), 50k (SAC)
DrawerOpen-v3	200	2M	DDPG, TD3	10k
DrawerClose-v3	200	2M	DDPG, TD3	10k

Table 3: Primary task and cohort definitions. Every method within a task–backbone cohort uses the same environment-step budget and exact evaluation grid. Primary comparisons use paired seeds 1–5.

Setting	Value or convention
HER	Future-goal convention in Sec. 1.2; nominal $p_{\mathrm{HER}} = 0.8$
Replay	Uniform for HER and SABER; capacity 300k; minibatch size 256
Networks	Actor and critic MLPs with two hidden layers of 256 units
Optimization	Adam, actor and critic learning rates 3×10^{-4}
Discount and targets	$\gamma = 0.98$, Polyak coefficient $\tau = 0.005$
Update schedule	UTD 1; learning begins after 5k steps; random actions for first 10k steps
Termination	Successful relabeled transitions are nonterminal; time limits bootstrap
Evaluation	Deterministic policy, 20 episodes per checkpoint
Statistics	Seed mean and sample SD; paired percentile-bootstrap CI, 10,000 resamples, RNG seed 2027
Curve processing	Exact common checkpoints only; no smoothing, interpolation, extrapolation, or filling

Table 4: Shared training, evaluation, and statistical settings. Matched HER–SABER runs use identical replay and relabeling code.

Backbone	Critic	Exploration or entropy	Backbone-specific settings
DDPG	One critic	Gaussian action noise, SD 0.2	Deterministic actor
TD3	Twin critics	Gaussian action noise, SD 0.2	Target noise 0.2, clip 0.5, policy delay 2
SAC	Twin critics	Automatic entropy tuning	Initial $\alpha = 0.2$, α LR 3×10^{-4}, target entropy -4

Table 5: Backbone-specific settings. SABER preserves each backbone's target construction and actor-update rule.

2.2 Comparison Methods

Method	Selection changed	Loss coefficient	Correction
HER	None beyond uniform replay and HER relabeling	Uniform	None
SABER	None	Current-policy local-support weight	Unit minibatch mean
HGR	Episode and transition–future-goal pair	HGR importance weight	Two-stage IS
PER	Replay transition	PER importance weight	Annealed IS
TD-error weighting	None	TD-error-derived weight	Unit minibatch mean

Table 6: Operational intervention points. HGR uses priority exponents $\alpha_{\mathrm{episode}} = \alpha_{\mathrm{pair}} = 0.6$, while PER uses $\alpha = 0.6$. Both anneal β linearly from 0.4 to 1.0 over critic-update progress and use priority offset 10^{-6}. TD-error weighting retains uniform replay and is therefore the closest allocation-stage control.

Prioritization controls. HGR prioritizes an episode and then a transition–future-goal pair, multiplies the two normalized importance factors on relabeled samples, and leaves the original-goal coefficient at one. PER samples replay transitions with TD-error priority and annealed importance correction. Both change realized selection, unlike SABER and the TD-error critic-loss weighting control.

TD-error critic-loss weighting. This control preserves uniform replay and changes only normalized per-sample critic coefficients. DDPG uses its single absolute TD error. SAC uses $\min(|\delta_{i1}|, |\delta_{i2}|)$ as the signal and applies the resulting coefficient to both critic losses. TD3 results are not reported because the stored control used independent target-policy smoothing noise for weight construction and optimized residuals, preventing the intended matched-signal interpretation.

2.3 Metrics and Statistical Analysis

Let $q_j \in [0, 1]$ be deterministic evaluation success at exact shared checkpoint t_j. Normalized success AUC is

$$\mathrm{AUC} = \frac{1}{t_m - t_1} \sum_{j=1}^{m-1} \frac{q_j + q_{j+1}}{2} (t_{j+1} - t_j).$$

Final-quarter AUC applies the same calculation to 750k–1M for FPP and 1.5M–2M for Drawer tasks. Final success is the value at the final checkpoint. Hit90 is the first exact checkpoint with $q_j \geq 0.9$; its median is computed only over seeds that reach the threshold, so reached-seed coverage is always reported.

For paired comparisons, each seed contributes one method difference. The reported 95% intervals are percentile intervals from 10,000 paired bootstrap resamples with pseudorandom seed 2027. Sample standard deviations use $n - 1$ in the denominator. No curve is smoothed, interpolated, extrapolated, or forward-filled.

Where reported, Steps/s is the arithmetic mean over seeds 1–5 of each run's completed environment steps divided by total training runtime. These historical measurements were not hardware-normalized across run families and are therefore descriptive implementation measurements, not learning-efficiency claims. Control and reference tables use the common order Cohort, Method/Variant, AUC, FQ-AUC, Final, Hit90, and Steps/s; diagnostic tables retain signal-specific columns.

3 Primary Comparison Details

3.1 Drawer Learning Curves

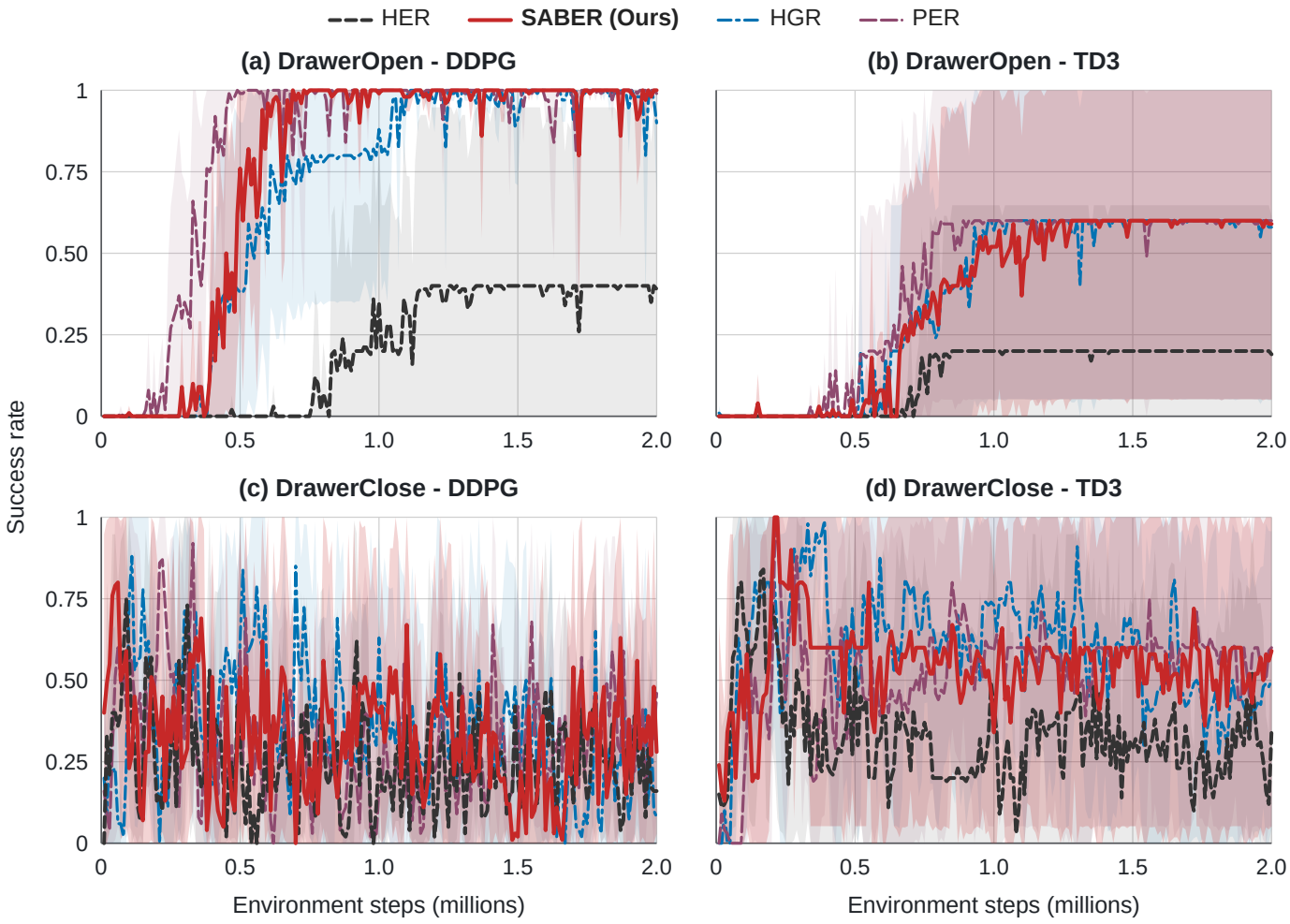


Figure 1: DrawerOpen and DrawerClose learning curves for DDPG and TD3. Lines are seed means and bands are sample standard deviations over paired seeds 1–5 on the exact 10k grid. No smoothing or interpolation is applied.

3.2 Time to Threshold

Hit90 is a threshold-crossing summary rather than a stability measure. Coverage is especially important when only a subset of seeds reaches 0.9.

Task	Backbone	Method	Median Hit90	Reached seeds	Steps/s
FetchPickAndPlace	DDPG	SABER	540k	5/5	41.7
		HER	600k	5/5	65.0
		HGR	930k	2/5	16.9
		PER	640k	4/5	58.0
FetchPickAndPlace	TD3	SABER	840k	5/5	41.2
		HER	980k	3/5	65.3
		HGR	630k	1/5	12.9
		PER	1.00M	1/5	50.8
FetchPickAndPlace	SAC	SABER	750k	5/5	29.1
		HER	800k	3/5	38.9
		HGR	800k	2/5	8.4
		PER	550k	5/5	34.8
DrawerOpen	DDPG	SABER	500k	5/5	23.3
		HER	960k	2/5	41.1
		HGR	530k	5/5	13.2
		PER	330k	5/5	34.0
DrawerOpen	TD3	SABER	780k	3/5	21.6
		HER	740k	1/5	41.5
		HGR	760k	3/5	11.6
		PER	720k	3/5	34.6
DrawerClose	DDPG	SABER	20k	5/5	24.0
		HER	70k	5/5	41.2
		HGR	100k	5/5	13.2
		PER	60k	5/5	33.9
DrawerClose	TD3	SABER	80k	5/5	22.5
		HER	70k	5/5	41.7
		HGR	80k	5/5	11.8
		PER	120k	5/5	34.4

Table 7: Hit90 for the primary comparisons. Hit90 is the first exact-grid checkpoint at which deterministic evaluation success reaches at least 0.9. Medians use only reached seeds, so reached-seed coverage is reported separately. Steps/s is the descriptive seed mean defined in Section 1.3.

3.3 FPP TD3 Confirmatory and HER-Indexing Sensitivity Results

The ten-seed cohort extends the frozen primary FPP TD3 comparison. The inclusive-future sensitivity also admits $m(s_{t+1})$, while leaving the matched HER–SABER difference otherwise unchanged. Its positive observed mean direction does not constitute an equivalence test between HER indexing conventions.

Seed	HER AUC	SABER AUC	Difference
1	0.3465	0.4301	$+0.0836$
2	0.3727	0.4568	$+0.0841$
3	0.3942	0.4960	$+0.1018$
4	0.4109	0.4066	-0.0043
5	0.3338	0.4225	$+0.0886$
6	0.4301	0.5207	$+0.0907$
7	0.3995	0.4614	$+0.0619$
8	0.3068	0.3543	$+0.0475$
9	0.4106	0.4114	$+0.0008$
10	0.4210	0.3331	-0.0879
Mean	0.3826	0.4293	$+0.0467$

Table 8: Frozen FPP TD3 confirmatory extension over ten paired seeds. AUC is computed on the exact 10k–1M evaluation grid without smoothing, interpolation, or filling. The paired mean difference is $+0.0467$ (10,000-resample paired-bootstrap 95% CI $[0.0085, 0.0783]$); 8/10 paired differences are positive. The exact two-sided sign-flip test over all 1,024 assignments gives $p = 0.0527$.

Seed	HER AUC	SABER AUC	Difference
1	0.3005	0.4576	$+0.1571$
2	0.2674	0.2669	-0.0005
3	0.3467	0.5179	$+0.1712$
4	0.3487	0.4003	$+0.0515$
5	0.3669	0.3247	-0.0422
Mean	0.3261	0.3935	$+0.0674$

Table 9: FPP TD3 sensitivity to inclusive future-goal indexing, which also admits $m(s_{t+1})$. AUC uses the exact 10k–1M evaluation grid. The paired mean difference is $+0.0674$ (10,000-resample paired-bootstrap 95% CI $[-0.0068, 0.1416]$), with 3/5 positive pairs and an exact two-sided sign-flip test of $p = 0.25$. The final-quarter AUC difference is $+0.1854$ (95% CI $[-0.0212, 0.3920]$), and the final-success difference is $+0.2200$ (95% CI $[-0.2000, 0.5200]$).

4 Additional Controls and Sensitivities

4.1 Allocation Signal

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s	ΔAUC
FPP–DDPG	SABER (Main)	0.609 ± 0.038	0.961 ± 0.014	0.960 ± 0.042	5/5 (0.540M)	41.7	$--$
	TD-error weighting	0.568 ± 0.069	0.910 ± 0.119	0.930 ± 0.104	5/5 (0.630M)	61.1	$+0.041$ $[-0.018, 0.123]$
DrawerOpen–DDPG	SABER (Main)	0.754 ± 0.044	0.987 ± 0.018	1.000 ± 0.000	5/5 (0.500M)	23.3	$--$
	TD-error weighting	0.715 ± 0.090	0.988 ± 0.011	0.800 ± 0.447	5/5 (0.610M)	39.0	$+0.038$ $[-0.015, 0.091]$
DrawerClose–DDPG	SABER (Main)	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	24.0	$--$
	TD-error weighting	0.313 ± 0.098	0.351 ± 0.218	0.400 ± 0.166	5/5 (0.130M)	38.9	$+0.011$ $[-0.079, 0.074]$
FPP–SAC	SABER (Main)	0.583 ± 0.030	0.986 ± 0.011	0.990 ± 0.022	5/5 (0.750M)	29.1	$--$
	TD-error weighting	0.544 ± 0.068	0.901 ± 0.167	0.990 ± 0.022	5/5 (0.800M)	38.2	$+0.040$ $[-0.016, 0.095]$

Table 10: Validated TD-error critic-loss weighting controls over paired seeds 1–5. Replay remains uniform, and only critic-loss coefficients change. Values are mean ± sample SD. The final column reports SABER-minus-control paired AUC differences with 95% bootstrap CIs; positive values favor SABER. For SAC, the priority signal is $\min(|\delta_{i1}|, |\delta_{i2}|)$, and the resulting normalized coefficient is applied to both critic losses. Steps/s is the descriptive seed mean defined in Section 1.3.

Across the validated DDPG and SAC cohorts in Table 10, SABER has higher observed mean AUC than TD-error critic-loss weighting. These controls isolate the weighting signal more closely than PER because both retain uniform replay.

4.2 Action-Bin Resolution

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s	Support+	ΔAUC
FPP–TD3	$K = 3$	0.353 ± 0.048	0.603 ± 0.121	0.750 ± 0.265	2/5 (0.945M)	33.9	0.396 ± 0.012	$+0.089$ $[0.070, 0.109]$
	$K = 5$ (Main)	0.442 ± 0.035	0.804 ± 0.103	0.970 ± 0.045	5/5 (0.840M)	41.2	0.219 ± 0.009	$--$
	$K = 7$	0.361 ± 0.047	0.616 ± 0.151	0.690 ± 0.263	2/5 (0.890M)	36.1	0.148 ± 0.010	$+0.081$ $[0.067, 0.093]$

Table 11: FPP TD3 action-bin sensitivity over paired seeds 1–5. Values are mean ± sample SD. Support+ is the support-positive rate. The final column reports Main-minus-variant paired AUC differences with 10,000-resample percentile-bootstrap 95% CIs; positive values favor Main $K = 5$.

Both $K = 3$ and $K = 7$ lower FPP TD3 mean AUC relative to the Main $K = 5$ setting in this paired sensitivity. The accompanying support rates show that bin resolution also changes activation frequency; this experiment does not isolate bin geometry from support

density.

4.3 Actor and Critic Weight Scope

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s	ΔAUC (Main–variant)
FPP–DDPG	Critic-Only (Main)	0.609 ± 0.038	0.961 ± 0.014	0.960 ± 0.042	5/5 (0.540M)	41.7	$--$
	Actor-Only	0.589 ± 0.032	0.952 ± 0.016	0.960 ± 0.022	5/5 (0.600M)	15.8	$+0.020[-0.024, 0.063]$
	Actor+Critic			*pending*			
DrawerOpen–DDPG	Critic-Only (Main)	0.754 ± 0.044	0.987 ± 0.018	1.000 ± 0.000	5/5 (0.500M)	23.3	$--$
	Actor-Only	0.612 ± 0.343	0.760 ± 0.433	0.800 ± 0.447	4/5 (0.470M)	18.6	$+0.141[-0.051, 0.478]$
	Actor+Critic	0.728 ± 0.084	0.992 ± 0.008	0.990 ± 0.022	5/5 (0.540M)	18.4	$+0.026[-0.048, 0.091]$
DrawerClose–DDPG	Critic-Only (Main)	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	24.0	$--$
	Actor-Only	0.307 ± 0.039	0.299 ± 0.061	0.280 ± 0.299	5/5 (0.040M)	24.0	$+0.017[-0.026, 0.078]$
	Actor+Critic	0.298 ± 0.071	0.241 ± 0.119	0.440 ± 0.370	5/5 (0.050M)	23.7	$+0.026[-0.043, 0.091]$

Table 12: DDPG actor-scope ablations over paired seeds 1–5. Values are mean ± sample standard deviation. Hit90 reports the number of reaching seeds and the reached-seed median step. Steps/s is the descriptive seed mean defined in Section 1.3. The final column gives Main-minus-variant paired AUC differences with 95% bootstrap confidence intervals. Pending cells are left unreported until a complete five-seed cohort is available.

Scope variants hold the support-query rule, support hyperparameters, and minibatch normalization fixed. Critic-Only applies Ω_i only to Bellman losses; Actor-Only applies it to per-sample actor terms; Actor+Critic applies it to both. Because each scope induces its own training trajectory, realized support indicators are not assumed to be identical across runs.

4.4 Secondary Replay and Update-Budget References

For both REDQ-HER TQB references, TQB denotes target-Q bounding: the sampled target-critic backup is clipped to the sparse-reward range $[-50, 0]$ before the Bellman target is formed. The no-HER AER comparisons test a different setting from the paper's post-HER question. The high-replay-ratio and REDQ-HER results demonstrate that additional updates and a larger critic ensemble can be stronger on FPP SAC, but they change UTD or architecture and therefore do not replace the matched UTD-1 comparison. On DrawerClose, all REDQ-HER seeds reach Hit90, but its mean final-quarter AUC is 0.098 and mean final success is 0.100, compared with 0.277 and 0.280 for DDPG SABER. This reference exhibits strong early threshold crossing without retaining the same success level late in training; because algorithm, UTD, and critic architecture all differ, it is not interpreted as a one-factor comparison with SABER.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s
FPP–DDPG	ER	0.148 ± 0.039	0.354 ± 0.124	0.460 ± 0.188	0/5	68.4
	Annealing ER	0.129 ± 0.111	0.259 ± 0.251	0.340 ± 0.363	1/5 (0.960M)	68.0
	AER	0.044 ± 0.002	0.045 ± 0.004	0.000 ± 0.000	0/5	67.0
FPP–TD3	ER	0.039 ± 0.002	0.043 ± 0.002	0.000 ± 0.000	0/5	21.0
	Annealing ER	0.039 ± 0.002	0.043 ± 0.002	0.000 ± 0.000	0/5	19.6
	AER	0.039 ± 0.002	0.043 ± 0.002	0.000 ± 0.000	0/5	26.2
FPP–SAC	ER	0.064 ± 0.042	0.114 ± 0.135	0.130 ± 0.199	0/5	39.4
	Annealing ER	0.041 ± 0.002	0.047 ± 0.008	0.000 ± 0.000	0/5	39.5
	AER	0.040 ± 0.003	0.044 ± 0.002	0.000 ± 0.000	0/5	39.3

Table 13: FPP no-HER replay references over seeds 1–5. Values are mean ± sample SD on the exact 10k–1M grid. These runs omit hindsight relabeling and are context references rather than matched post-HER mechanism controls. Steps/s is the descriptive seed mean defined in Section 1.3.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s
FPP–SAC	HER UTD 1; 2 critics	0.471 ± 0.079	0.735 ± 0.247	0.790 ± 0.288	3/5 (0.800M)	38.9
	SABER UTD 1; 2 critics	0.583 ± 0.030	0.986 ± 0.011	0.990 ± 0.022	5/5 (0.750M)	29.1
	HER high-RR UTD 4; 2 critics	0.768 ± 0.097	0.963 ± 0.021	0.960 ± 0.042	5/5 (0.450M)	5.9
	REDQ-HER TQB					
FPP–REDQ	UTD 4; 10 critics/2 target	0.867 ± 0.052	0.993 ± 0.008	1.000 ± 0.000	5/5 (0.300M)	3.1

Table 14: FPP SAC and REDQ scaling references on the exact shared 50k–1M grid. Values are mean ± sample SD over seeds 1–5. High-replay-ratio HER and REDQ-HER change the update budget, and REDQ-HER also changes critic architecture; they are not matched post-HER loss-allocation controls. Steps/s is the descriptive seed mean defined in Section 1.3.

Cohort	Method/Variant	AUC	FQ-AUC	Final	Hit90	Steps/s
DrawerClose–DDPG	HER UTD 1; 1 critic	0.251 ± 0.030	0.250 ± 0.093	0.160 ± 0.182	5/5 (0.070M)	41.2
	SABER UTD 1; 1 critic	0.324 ± 0.082	0.277 ± 0.100	0.280 ± 0.251	5/5 (0.020M)	24.0
	REDQ-HER TQB					
DrawerClose–REDQ	UTD 4; 10 critics/2 target	0.188 ± 0.041	0.098 ± 0.022	0.100 ± 0.117	5/5 (0.020M)	5.8

Table 15: DrawerClose update-budget and architecture reference over seeds 1–5. Values are mean ± sample standard deviation on the exact 10k–2M grid. FQ-AUC covers 1.5M–2M; Final is final-checkpoint success; Hit90 gives reaching seeds and their median first-hit step. REDQ-HER uses four updates per environment step and a 10-critic ensemble with two target critics sampled per backup, so it is not a resource-matched SABER comparison. Steps/s is the descriptive seed mean defined in Section 1.3.

5 Support-Signal Diagnostic Details

5.1 Instrumentation and Aggregation

Diagnostic logging introduces no gradient term and does not change replay sampling, objectives, or parameter-update order. FPP logs one full minibatch every 10k environment steps for DDPG, TD3, and SAC seeds 1–5: 100 batches per run, 500 per backbone, and 1,500 total. Drawer diagnostics use a 200-step cadence from 5k through 2M, producing 9,976 persisted snapshots per seed. Snapshot quantities are aggregated within seed before cohort statistics are computed.

For each persisted minibatch, the diagnostic stores the pre-update absolute Bellman residual, support membership, and the residual after one critic update. The fixed-target residual is

$$\Delta|\delta_i| = \left|Q_\phi^{\mathrm{pre}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right| - \left|Q_\phi^{\mathrm{post}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right|.$$

Thus the target and minibatch are held fixed while the critic parameters change. Live-target quantities, where available, are retained only for audit and are not mixed with the fixed-target statistic.

5.2 Cohort-Level Results

Top-10% enrichment is the support-positive rate among the exact $\lceil 0.1|B| \rceil$ samples with largest pre-update absolute residual, divided by the full-minibatch support-positive rate. Stable descending order breaks ties, so the selected set has exactly the intended size. The modest Pearson correlations and enrichment above one show that support and TD error are related but not identical rankings. For DrawerOpen, fixed-target residual reduction is larger for support-positive samples in all five primary seeds and all five extension seeds.

5.3 Phase-Conditioned Allocation

For both Drawer tasks, the primary seeds and the extensions show higher support activation in approach and contact than in the dominant terminal phase. This phase analysis describes where the indicator activates; it is not an independent causal test of

Cohort	Seeds	Support+ rate	Pearson r	Top-10% enrich.	Pre-update $\lvert\delta\rvert$ $(+/-)$	One-step $\Delta\lvert\delta\rvert$ $(+/-)$
FPP–DDPG	1–5	0.367 ± 0.011	0.116	$1.409\times$	$1.42\times$	$2.39\times$
FPP–TD3	1–5	0.219 ± 0.009	0.128	$1.670\times$	$1.62\times$	$3.30\times$
FPP–SAC	1–5	0.197 ± 0.013	0.143	$1.828\times$	$1.74\times$	$3.94\times$
DrawerOpen–DDPG	1–5	0.417 ± 0.021	0.084	$1.379\times$	$1.16\times$	$1.79\times$
DrawerOpen–DDPG	6–10 extension	0.475 ± 0.094	0.095	$1.440\times$	$1.13\times$	$1.91\times$
DrawerOpen–DDPG	1–10 pooled	0.446 ± 0.071	0.090	$1.409\times$	$1.15\times$	$1.83\times$
DrawerClose–DDPG	1–5	0.298 ± 0.018	0.104	$1.534\times$	$1.42\times$	$2.94\times$
DrawerClose–DDPG	6–10 extension	0.299 ± 0.007	0.118	$1.602\times$	$1.48\times$	$4.23\times$
DrawerClose–DDPG	1–10 pooled	0.298 ± 0.013	0.111	$1.568\times$	$1.45\times$	$3.48\times$

Table 16: Support-signal diagnostics. Statistics are first aggregated within seed, after which seeds are independent units. Support rates are mean $\pm$ sample SD. The last two columns divide support-positive by support-negative cohort means. One-step $\Delta\lvert\delta\rvert$ evaluates the same minibatch after one critic step while retaining the pre-update Bellman target. Seeds 1–5 are the primary diagnostic cohorts, seeds 6–10 are separate extensions, and pooled ten-seed rows are descriptive.

Cohort	Phase	Sample fraction	Support+ rate	Seeds
DrawerOpen–DDPG	Approach	0.174 ± 0.018	0.745 ± 0.022	1–5
	Contact	0.049 ± 0.003	0.698 ± 0.056	
	Opening	0.046 ± 0.005	0.636 ± 0.053	
	Open	0.730 ± 0.018	0.306 ± 0.027	
DrawerOpen–DDPG	Approach	0.195 ± 0.011	0.754 ± 0.030	6–10 ext.
	Contact	0.027 ± 0.024	0.515 ± 0.181	
	Opening	0.023 ± 0.023	0.440 ± 0.263	
	Open	0.755 ± 0.039	0.387 ± 0.127	
DrawerClose–DDPG	Approach	0.218 ± 0.002	0.549 ± 0.032	1–5
	Contact	0.016 ± 0.003	0.710 ± 0.026	
	Closing	0.019 ± 0.004	0.425 ± 0.034	
	Closed	0.747 ± 0.004	0.212 ± 0.017	
DrawerClose–DDPG	Approach	0.222 ± 0.003	0.569 ± 0.004	6–10 ext.
	Contact	0.018 ± 0.003	0.727 ± 0.016	
	Closing	0.017 ± 0.002	0.438 ± 0.028	
	Closed	0.743 ± 0.007	0.204 ± 0.010	

Table 17: Drawer phase-conditioned allocation. Phases are defined from the relabeled training goal. Values are seed means $\pm$ sample SD; support is sample-weighted within each seed. Seeds 6–10 are separate extensions. Pooled ten-seed and seed-level values are included in the artifacts.

long-horizon performance.

6 Runtime and Resource Measurements

6.1 Execution-Environment Audit

Host	System	Compute hardware	OS, kernel, and driver
A	MSI GB10 system	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.4; 6.17.0-1014-nvidia; driver 580.142
B	GIGABYTE AI TOP ATOM	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.4; 6.17.0-1026-nvidia; driver 580.159.03
C	GIGABYTE AI TOP ATOM	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.4; 6.17.0-1026-nvidia; driver 580.159.03
D	GB10 system	20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10	Ubuntu 24.04.3; 6.11.0-1016-nvidia; driver 580.95.05

Table 18: Hardware inventory for the four physically distinct execution hosts. All systems use CUDA-addressable unified memory rather than dedicated VRAM. Hosts B and C are separate machines with the same model and system configuration. Hostnames, serial numbers, and other identifying metadata are omitted.

All four hosts used the verified common experiment-library stack: Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.20.0, Gymnasium 1.3.0, Gymnasium-Robotics 1.4.2, MuJoCo 3.3.0, Meta-World 3.1.1, NumPy 2.3.5, and SciPy 1.18.0. For each paired HER–SABER cohort, both methods and all paired seeds were executed on one physical host under this stack; no paired cohort was split across machines. Hardware differs across cohorts, so throughput values remain descriptive and are not interpreted across hosts. Sanitized run-family and source-provenance records are provided in `execution_host_mapping.csv`.

6.2 Dedicated Runtime Profile

The dedicated profile sequentially executes HER and SABER on the recorded stack, using one seed and the 20k–100k interval. It supports attribution of implementation overhead but not a claim about asymptotic complexity or end-to-end learning efficiency. Historical throughput values in the result tables were not collected under this dedicated sequential profiling protocol; they are therefore descriptive rather than compute-matched. Peak memory was not recorded and is reported as not measured rather than estimated.

Profile item	Recorded value
Execution protocol	Sequential FPP TD3 HER and SABER, seed 1, 20k–100k
GPU	NVIDIA GB10, compute capability 12.1, 121.7 GiB CUDA-addressable memory
CPU/platform	20 logical CPUs, AArch64 Linux 6.17.0-1026-nvidia, glibc 2.39
Software	Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.20.0
Comparison boundary	The dedicated profile is not extrapolated to historical primary runs
Memory boundary	Peak process and accelerator memory were not measured

Table 19: Recorded infrastructure for the dedicated same-hardware runtime profile. CUDA-addressable memory is reported because the GB10 uses unified memory. These identifiers characterize the dedicated runtime profile; historical steps/s values in other tables are descriptive cohort measurements.

Component	HER ms/step	SABER ms/step	Difference	SABER share
Interaction	1.000	1.071	+0.070	4.4%
Replay + HER	1.925	2.651	+0.726	11.0%
Batch preparation	0.096	0.108	+0.012	0.4%
Support and weighting	0.000	15.028	+15.028	62.3%
Critic update	2.945	2.832	−0.113	11.7%
Actor + target update	1.384	1.292	−0.091	5.4%
Evaluation/logging amortized	0.080	0.085	+0.005	0.4%
Other loop overhead	0.633	1.066	+0.434	4.4%
Total training loop	8.063	24.133	+16.070	100.0%

Table 20: Hardware-specific FPP TD3 implementation profile over the 20k–100k window of seed 1. HER and SABER achieve 124.0 and 41.4 environment steps/s excluding evaluation, respectively. The profile identifies current support retrieval as the dominant overhead. Other loop overhead is the difference between total loop time and the explicitly instrumented components. The profile is not a learning-performance claim and is not extrapolated to 1M steps.

Task	Backbone	HGR	SABER	Ratio
FPP	DDPG	16.93	41.72	2.46×
FPP	TD3	12.94	41.25	3.19×
FPP	SAC	8.39	29.10	3.47×
DrawerOpen	DDPG	13.24	23.33	1.76×
DrawerOpen	TD3	11.65	21.57	1.85×
DrawerClose	DDPG	13.22	23.96	1.81×
DrawerClose	TD3	11.81	22.49	1.90×

Table 21: Recorded environment steps/s for HGR and SABER. Values are descriptive means over seeds 1–5 within each cohort. These runs were not collected under the dedicated sequential profiling protocol, so the values support descriptive within-cohort comparisons and no cross-cohort throughput ranking.

7 Artifact and Provenance Index

Content	Artifact key	Status
Primary seed metrics and curves	`primary_seed_results`	Complete
Action-bin sensitivity	`action_bin_sensitivity`	Complete
DDPG actor-scope analysis	`actor_scope_ddpg`	FPP A+C pending
Derived Technical Supplement summaries	`technical_supplement`	FPP A+C pending
FPP diagnostics	`fpp_diagnostic`	Complete
DrawerClose diagnostics	`drawerclose_diagnostic`	Complete
DrawerOpen fixed-target diagnostics	`draweropen_fixed_target`	Complete
FPP TD3 runtime profile	`fpp_td3_runtime`	Complete
Execution-environment audit	`execution_environment_audit`	Partial
FPP TD3 ten-seed confirmation	`fpp_td3_n10`	Complete
FPP TD3 inclusive-future sensitivity	`fpp_td3_inclusive`	Complete
FPP SAC update-budget references	`fpp_sac_scaling`	Complete
DrawerClose REDQ scaling reference	`drawerclose_redq`	Complete

Table 22: Artifact index. Keys resolve to repository-relative package paths in `artifact_index.csv`. Canonical artifacts include manifests that bind source files by repository-relative path and SHA256.

Development scope and seed mapping. We did not conduct an exhaustive joint hyperparameter search. The Main SABER settings were fixed for the matched comparisons, and no control received variant-specific tuning. The $K \in \{3, 7\}$ runs are sensitivity checks around Main $K = 5$, not a per-task model-selection sweep. For paired seed s, the environment, PyTorch, NumPy, and action-noise

streams use s; the support-weight stream uses $20000 + s$, and evaluation uses $10000 + s$. Training and evaluation resets advance by deterministic episode- and checkpoint-indexed offsets. All methods in a paired cohort use this mapping.

Generated analysis directories contain compact CSV summaries, a repository-relative source manifest, and a self-excluding artifact manifest. Canonical experiment artifacts retain configuration, environment snapshot, source provenance, and validation records where available. Raw result files are referenced rather than duplicated in the derived summaries.

A Primary Seed-Level Results

The following tables report every primary seed value used in the aggregate comparison. AUC and final-quarter AUC use each cohort's exact common grid.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.640152	0.956000	0.900	0.510M
	2	0.537374	0.934000	0.950	0.600M
	3	0.578788	0.964000	0.950	0.610M
	4	0.325253	0.731000	0.900	0.920M
	5	0.612374	0.967000	1.000	0.570M
SABER	1	0.656061	0.939000	0.950	0.480M
	2	0.615152	0.972000	0.950	0.540M
	3	0.564899	0.959000	1.000	0.630M
	4	0.631818	0.976000	1.000	0.480M
	5	0.577525	0.960000	0.900	0.610M
HGR	1	0.441667	0.765000	1.000	0.870M
	2	0.076515	0.065000	0.000	–
	3	0.374495	0.482000	0.500	–
	4	0.417677	0.587000	0.700	0.990M
	5	0.380556	0.506000	0.500	–
PER	1	0.484091	0.919000	1.000	0.770M
	2	0.520455	0.953000	1.000	0.730M
	3	0.604293	0.946000	0.900	0.550M
	4	0.643434	0.964000	1.000	0.460M
	5	0.049242	0.046000	0.000	–

Table 23: Seed-level primary results for FPP with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.346465	0.523000	0.550	–
	2	0.372727	0.635000	0.950	1.000M
	3	0.394192	0.662000	1.000	0.980M
	4	0.410859	0.686000	0.950	0.970M
	5	0.333838	0.530000	0.750	–
SABER	1	0.430051	0.846000	1.000	0.840M
	2	0.456818	0.856000	1.000	0.860M
	3	0.495960	0.914000	0.950	0.630M
	4	0.406566	0.652000	0.900	0.630M
	5	0.422475	0.752000	1.000	0.910M
HGR	1	0.345202	0.515000	0.500	–
	2	0.396212	0.593000	0.800	–
	3	0.407576	0.581000	0.850	–
	4	0.326515	0.489000	0.450	0.630M
	5	0.361111	0.493000	0.450	–
PER	1	0.345455	0.515000	0.550	–
	2	0.408586	0.653000	0.950	1.000M
	3	0.394192	0.628000	0.800	–
	4	0.369192	0.526000	0.500	–
	5	0.327020	0.488000	0.500	–

Table 24: Seed-level primary results for FPP with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.559211	0.985000	1.000	0.800M
	2	0.469737	0.760000	1.000	0.950M
	3	0.406579	0.500000	0.500	–
	4	0.540789	0.965000	1.000	0.800M
	5	0.378947	0.465000	0.450	–
SABER	1	0.636842	0.995000	0.950	0.650M
	2	0.567105	0.980000	1.000	0.800M
	3	0.573684	0.980000	1.000	0.750M
	4	0.575000	1.000000	1.000	0.750M
	5	0.564474	0.975000	1.000	0.800M
HGR	1	0.451316	0.545000	0.800	–
	2	0.596053	0.980000	1.000	0.700M
	3	0.506579	0.800000	1.000	0.900M
	4	0.450000	0.505000	0.500	–
	5	0.414474	0.490000	0.500	–
PER	1	0.755263	0.970000	1.000	0.450M
	2	0.635526	0.985000	1.000	0.550M
	3	0.706579	0.990000	1.000	0.550M
	4	0.690789	1.000000	1.000	0.550M
	5	0.660526	0.985000	0.950	0.600M

Table 25: Seed-level primary results for FPP with SAC. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.460302	0.970000	1.000	1.090M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000000	0.000000	0.000	–
	5	0.576256	0.993500	0.950	0.830M
SABER	1	0.757286	1.000000	1.000	0.510M
	2	0.786935	0.977000	1.000	0.420M
	3	0.705025	1.000000	1.000	0.580M
	4	0.713065	0.960000	1.000	0.500M
	5	0.806030	1.000000	1.000	0.400M
HGR	1	0.799749	0.995000	0.500	0.410M
	2	0.704523	0.990000	1.000	0.530M
	3	0.692211	0.993000	1.000	0.610M
	4	0.794975	0.998000	1.000	0.430M
	5	0.449497	0.934500	1.000	1.060M
PER	1	0.859799	1.000000	1.000	0.330M
	2	0.877387	0.998000	1.000	0.270M
	3	0.814070	0.969000	1.000	0.330M
	4	0.809171	0.951500	0.950	0.390M
	5	0.765075	0.990000	1.000	0.410M

Table 26: Seed-level primary results for DrawerOpen with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.632286	0.999500	0.950	0.740M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.000251	0.000000	0.000	–
	5	0.000000	0.000000	0.000	–
SABER	1	0.486558	0.996500	0.950	1.070M
	2	0.000000	0.000000	0.000	–
	3	0.000503	0.000000	0.000	–
	4	0.689950	0.997000	1.000	0.560M
	5	0.632161	0.997000	1.000	0.780M
HGR	1	0.000000	0.000000	0.000	–
	2	0.692462	0.992000	1.000	0.520M
	3	0.531910	0.997000	0.900	0.930M
	4	0.000000	0.000000	0.000	–
	5	0.628266	0.996000	1.000	0.760M
PER	1	0.632412	0.986000	1.000	0.780M
	2	0.000000	0.000000	0.000	–
	3	0.000000	0.000000	0.000	–
	4	0.755528	0.999000	1.000	0.520M
	5	0.644221	0.994000	1.000	0.720M

Table 27: Seed-level primary results for DrawerOpen with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.281658	0.337500	0.300	0.090M
	2	0.268844	0.315500	0.100	0.150M
	3	0.203518	0.193000	0.400	0.040M
	4	0.249497	0.289000	0.000	0.070M
	5	0.251508	0.116000	0.000	0.020M
SABER	1	0.287814	0.165500	0.650	0.010M
	2	0.337437	0.417000	0.200	0.060M
	3	0.284045	0.306500	0.150	0.050M
	4	0.461307	0.300500	0.000	0.020M
	5	0.251508	0.197500	0.400	0.010M
HGR	1	0.672613	0.218000	0.000	0.100M
	2	0.323869	0.236500	0.000	0.090M
	3	0.237563	0.154500	0.150	0.120M
	4	0.238568	0.219500	0.050	0.100M
	5	0.273367	0.299000	0.200	0.010M
PER	1	0.288568	0.272000	0.550	0.100M
	2	0.244975	0.273500	0.000	0.180M
	3	0.297990	0.392000	1.000	0.040M
	4	0.331281	0.356500	0.350	0.050M
	5	0.376884	0.404000	0.400	0.060M

Table 28: Seed-level primary results for DrawerClose with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

Method	Seed	AUC	FQ-AUC	Final	Hit90 step
HER	1	0.450628	0.455000	0.850	0.080M
	2	0.058291	0.002000	0.000	0.120M
	3	0.886181	0.847000	0.900	0.050M
	4	0.097236	0.017000	0.000	0.060M
	5	0.151382	0.098000	0.000	0.070M
SABER	1	0.080653	0.020000	0.000	0.080M
	2	0.908291	0.978000	1.000	0.090M
	3	0.739950	0.761000	1.000	0.200M
	4	0.908920	0.909500	0.950	0.010M
	5	0.106030	0.000000	0.000	0.050M
HGR	1	0.923618	0.886000	0.400	0.080M
	2	0.686307	0.808500	0.950	0.060M
	3	0.782161	0.428000	1.000	0.130M
	4	0.477010	0.115000	0.150	0.060M
	5	0.138945	0.063000	0.000	0.260M
PER	1	0.111055	0.050000	0.000	0.220M
	2	0.845477	0.987000	1.000	0.190M
	3	0.824874	0.891000	1.000	0.120M
	4	0.112060	0.002000	0.000	0.120M
	5	0.725377	0.983000	1.000	0.100M

Table 29: Seed-level primary results for DrawerClose with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.